

Griffin Kuchar

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EDUCATION

Texas Christian University (TCU)

Expected May 2028

B.S. in Computer Science (May 2027) | Minors: Mathematics, Economics | GPA: 3.81/4.0

Fort Worth, TX

- **Continuing Education:** Directly into an M.S. in Computer Science (August 2027 – May 2028)
- **Relevant Coursework:** Data Structures & Algorithms, Database Systems, Operating Systems, Machine Learning, Deep Learning, Software Engineering, Web Technologies, GPU Computing, OOP & Design Patterns, Econometrics

EXPERIENCE

AMD

May 2026 – August 2026

Systems Engineering Intern | Mobile Systems (SoC) Team

Austin, TX

- Developed an agentic AI web application that automates first-level hardware debugging, reducing triage time by **80%** for **250+** validation and systems engineers; Python/FastAPI backend, React frontend, PostgreSQL, Redis.
- Designed the agent as an LLM-driven MCP & local tool-calling loop that autonomously runs triage and debug procedures on downloaded log files, iterating until it declares a root cause or reaches its token/loop limit.
- Implemented Jira OAuth 2.0 integration and AMD OIDC SSO, improving the end-user experience by reducing the number of steps to sign in by **40%**.

TCU NextGen AI Research Lab

January 2026 – Present

Machine Learning Research Assistant

Fort Worth, TX

- Engineered an end-to-end forecasting pipeline predicting pediatric patient decompensation **15** hours in advance, in collaboration with Children's Health Dallas and the TCU Nursing Department.
- Benchmarked 11 Machine Learning and Deep Learning classification models on 20,000+ anonymized patient records; optimized the top performer to reach 0.98 AUROC and **0.70** PR-AUC on a **1%** positive class dataset.

AMD

May 2025 – August 2025

Validation/Debug Engineering Intern | System Validation Team

Austin, TX

- Built a LLM chat UI adopted by **20+** engineers that queries Jira tickets by translating plain-English into structured JQL via the OpenAI API, cutting ticket-searching time by **50%**; Python/Flask backend, React frontend.
- Validated AMD CPUs/APUs by designing and executing stress-test workloads with 15+ unique components across 20+ systems. Triageed **100+** test failures and root-caused **~5** BIOS issues in platform-level systems.

PROJECTS

GPU-Accelerated CNN Emotion Detector | *Python, C++, CUDA, CuPy, NumPy, OpenCV, Streamlit* [GitHub](#)

- Trained a CNN emotion-classification model accelerated with custom CUDA kernels via CuPy on 36,000+ images across three emotion categories; reached 69% test accuracy and served inference through a Streamlit GUI.
- Achieved a **9.07×** speedup over its vectorized NumPy CPU implementation.

FrogCrew Event Scheduler | *Java (Spring Boot), REST APIs, JavaScript (Vue.js), Docker, Azure* [GitHub](#)

- Delivered a full-stack scheduling platform in Agile sprints for TCU Sports Broadcasting, streamlining event assignment for students and faculty across 7 sports and cutting scheduling time by **~70%**.
- Containerized the application with Docker and deployed to Azure, adding a GitHub Actions CI/CD pipeline that runs an automated test suite on every push.

Big-Tech Stock Classifier | *Python, Jupyter, scikit-learn, pandas, NumPy* [GitHub](#)

- Built a pipeline predicting the next day's best-performing tech stock using 6 years of OHLCV market data, custom technical indicators, and expanding-window time-series validation; achieved a **1.52×** lift over a random baseline.

LEADERSHIP & ACTIVITIES

Vice President, Upsilon Pi Epsilon (CS Honor Society), TCU Chapter

2025 – Present

Undergraduate Teaching Assistant, TCU Computer Science Department

2025 – 2026

Entrepreneur & Content Creator, YouTube (60K+ Subscribers)

2023 – 2026

TECHNICAL SKILLS

Languages: Python, C++, Java, C, Bash, TypeScript, JavaScript, HTML, CSS, SQL, R, CUDA

ML / AI: PyTorch, scikit-learn, pandas, NumPy, matplotlib, Jupyter, CuPy, LLM APIs (OpenAI, Claude), Tidyverse

Web / Backend: React, Next.js, Tailwind CSS, Node.js, FastAPI, Flask, Django, Spring Boot, Vue.js, Streamlit

Cloud / DevOps / Database: Git, Docker, Linux, AWS, Azure, nginx, PostgreSQL, MySQL, Oracle, Redis, Snowflake